

Abstract geometric lines in the top left corner, consisting of several overlapping, irregular polygons and lines in a light beige color, creating a modern, minimalist design.

UNSUPERVISED EXPLAINABLE IMAGE ANOMALY DETECTION

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EXECUTIVE SUMMARY

Motivation

Importance: anomaly detection in computer vision is incredibly important, especially in high-stakes industries like healthcare tumor identification and manufacturing quality control.

Challenges:

1. We rarely have enough labeled data for anomalies because defects are inherently rare and unpredictable.
2. Traditional black-box models lack of explainability required for human operators to actually trust them in production.

Main Idea

- Engineered an unsupervised, patch-based feature extraction pipeline.
- Leveraged a pre-trained ResNet-18 to flag defects mathematically via spatial distance, completely eliminating the need for labeled anomaly data.
- Natively produce a heatmap visual for explainability.

Result

- Ultimately, the pipeline successfully localized defects without ever training on them, achieving an ROC-AUC of **0.9350** and generating pixel-precise heatmaps without relying on traditional backward gradients

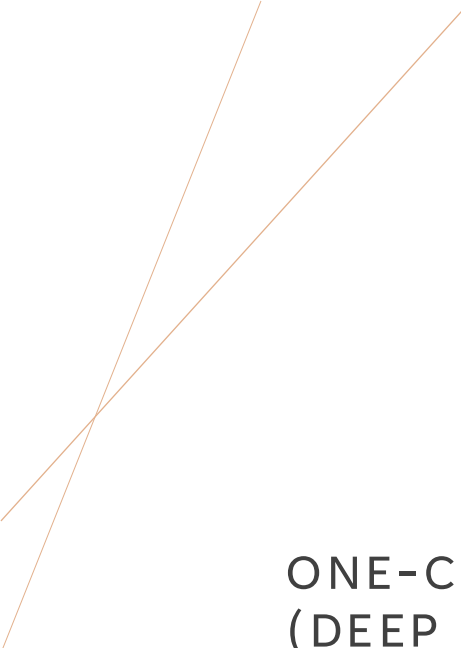
LITERATURE REVIEW

RECONSTRUCTION & GENERATIVE MODELS (AUTOENCODERS, GANS)

Mechanism: Trained to reconstruct pristine images. The underlying premise is that a network trained exclusively on normal images should theoretically fail to reconstruct a defect.

- Autoencoders: Autoencoders compress input images through a strict, low-dimensional mathematical bottleneck to filter out unseen features, flagging anomalies by measuring the pixel-wise reconstruction error between the defective input and the network's generated "normal" output.
- GANs: GANs perform an iterative search through a trained latent space to generate the closest possible "perfect" match to a test image, isolating anomalies by calculating the mathematical deviation between the real input and the generator's nominal approximation.

Limitation: However, these suffer from '**over-generalization.**' Deep models are often too powerful; they successfully reconstruct the anomaly itself, which reduces the mathematical error to zero and allows the defect to go undetected.



LITERATURE REVIEW

ONE-CLASS CLASSIFICATION (DEEP SVDD)

To solve the over-generalization problem, researchers shifted to **One-Class Classification**, like Deep SVDD.

Mechanism: Deep SVDD trains a neural network to map all normal data points into a tightly bound mathematical sphere, flagging anomalies whenever a test image's global feature vector falls outside this established boundary.

Limitation: **Loss of Spatial Context.** Standard models pool the entire image into a single global vector. It successfully flags a defective image, but cannot natively localize the defect.



LITERATURE REVIEW

FEATURE-BASED METHODS (THE CHOSEN ARCHITECTURE)

Mechanism: Maps local image patches into a nominal memory bank. During inference, anomalies are detected by calculating the direct Euclidean distance between the test features and the memory bank.

*Advantage: **Intrinsic Explainability.*** By calculating distance patch-by-patch, the spatial distance map inherently becomes the explanation, allowing for highly precise, native localization.

FEATURE EXTRACTION (VIA RESNET-18)

- Used pre-trained CNN, **ResNet-18**.
- Extract intermediate layers to capture both local textures and global structures (inspired by PaDiM).
- This gives us the perfect balance: it captures fine, microscopic textures as well as the broader structure of the object.

THE "NORMAL" MEMORY BANK

- Take all defect-free training images, pass them through the network
- Break them into spatial patches.
- Calculate their mathematical average - baseline of a 'perfect' image

DISTANCE-BASED INFERENCE

- Process new, unknown test images through the same network.
- Extract its patches and calculate the exact Euclidean distance between each test patch and the normal memory bank.

NATIVE EXPLAINABILITY

- Flag anomalies based on spatial distance thresholds.
- Map the patch distances directly into a high-resolution heatmap to visually isolate the defect. – Benefit of a patch-by-patch basis distance calculation

APPROACH

THE FEATURE-MEMORY PIPELINE

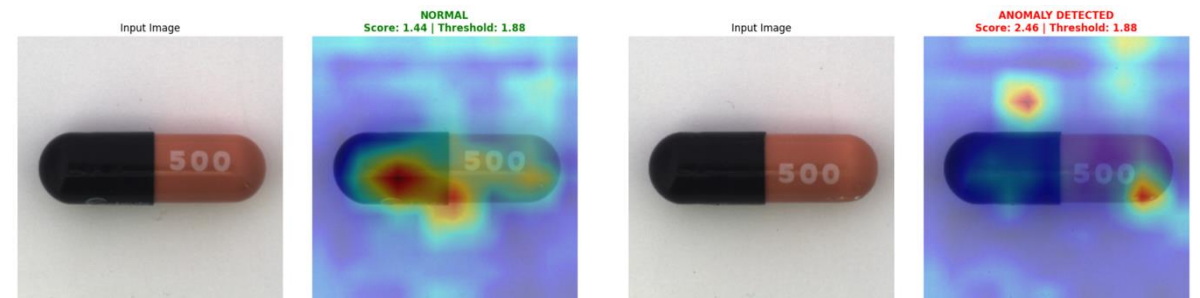
IMPLEMENTATION & DEMO

CODE IMPLEMENTATION

https://github.com/Allison1212/i-hw-Spring-2026-mini-research/blob/main/Code_demo_final.ipynb

SYSTEM DEMO

<https://youtu.be/5FdHAq7infk>



Two thin orange lines intersecting on the left side of the slide. One line is horizontal, and the other is diagonal, crossing it.

REFERENCE

- <https://arxiv.org/html/2302.06670v4>
- https://openreview.net/pdf?id=H1lK_lBtvS
- <https://arxiv.org/abs/2011.08785>
- <https://arxiv.org/abs/2106.08265>
- <https://www.mvtec.com/research-teaching/datasets/mvtec-ad>